

L	T	P	C
2	1	0	3

Course Code: INT422

IT PROJECT MANAGEMENT

Course Objectives:

This course will help the learner to understand the key aspects of software projects, the strategies required for managing projects from their birth to completion, agile project management techniques such as Scrum and DevOps.

UNIT - I

12 Periods

Project Overview and Feasibility Studies: Identification of Market and Demand Analysis, Project Cost Estimate, Financial Appraisal. **Project Scheduling:** Introduction to PERT and CPM - Critical Path Calculation - Precedence Relationship - Difference between PERT and CPM - Float Calculation and its importance - Cost reduction by Crashing of activity.

UNIT - II

11 Periods

Cost Control and Scheduling: Project Cost Control (PERT/Cost) - Resource Scheduling & Resource Leveling. **Project Management Features:** Risk Analysis - Project Control - Project Audit and Project Termination.

UNIT – III

11 Periods

Agile Project Management: Introduction to Agile Principles and Agile methodologies - Relationship between Agile Scrum, Lean, DevOps and IT Service Management (ITIL). **Scrum:** Various terminologies used in Scrum (Sprint, product backlog, sprint backlog, sprint review, retro perspective) - Various Scrum roles - Best practices of Scrum.

UNIT - IV

11 Periods

DevOps: Overview and its Components - Containerization Using Docker - Managing Source Code and Automating Builds - Automated Testing and Test Driven Development - Continuous Integration and Configuration Management - Continuous Deployment - Automated Monitoring. **Other Agile Methodologies:** Introduction to XP, FDD, DSDM & Crystal.

TEXTBOOKS

1. Mike Cohn, *Succeeding with Agile: Software Development Using Scrum*, Addison-Wesley, 2010
2. Rossberg, Joachim. *Agile Project Management with Azure DevOps: Concepts, Templates, and Metrics*. Germany, Apress, 2019.

REFERENCES

1. Roman Pichler, *Agile Product Management with Scrum*, Addison-Wesley Professional, 2010
2. Mark C. Layton, Steven J. Ostermiller, *Agile Project Management For Dummies*, Wiley, 2017
3. Rubin, Kenneth S. *Essential Scrum: A Practical Guide to the Most Popular Agile Process*. United Kingdom, Addison-Wesley, 2012.
4. Ken Schwaber, *Agile Project Management with Scrum (Microsoft Professional)*, Microsoft Press, 2004

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	- Compare and contrast the different methods of demand analysis and the impact of errors in operating cost forecast - Decide why and when to carry out a financial feasibility analysis and plan the project timeline using PERT and CPM
Unit II	- Accomplish situational assessment on resource levelling and crashing procedure for project completion - Develop proper audit and design project plan to foresee possible sources of risk and completion delay
Unit III	- Compare lean and scrum methodology principles - Implement agile software development using the scrum framework
Unit IV	- Discover a powerful automation DevOps platform - Implement DevOps project Management

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to:

CO No.	Course Outcome	Knowledge Level
1	Articulate feasibility study and frame project milestones considering critical activities using PERT and CPM	K2
2	Ensure time, cost and resource estimates are unbiased and reliable	K2
3	Develop an audit approach with regular review reports to ensure the assurance of project completion.	K3
4	Create and use scrum artifacts in agile project management	K3
5	Use Devops and Agile in tandem to streamline project development and management	K3